



Effect of Supplementation of Herbal Products as Feed Additive on Rumen Metabolites in Marwari Goats

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10.18805/ijar.B-3932

ABSTRACT

The present study was conducted to investigate the effect of herbal products supplementation on rumen environment in Marwari goats. Marwari goats of 18-24 months of age were randomly divided into five groups (four in each) i.e. T₁ (complete feed without herbal products), T₂ (Complete feed + Himalayan Batisa @ 10 mg/g substrate), T₃ (Complete feed+Appetonic Vet powder @ 10 mg/g substrate), T₄ (Complete feed + Ruchamax @ 3 mg/g substrate) and T₅ (Complete feed + Rumizyme powder @ 7.5 mg/g substrate). Rumen liquor samples were collected from each animal at 0, 2, 4, 6, 8, 12 and 18 hrs post prandial. Rumen pH was unaffected while TVFA concentration was higher (P<0.01) in herbal product supplemented groups in comparison to un-supplemented group. The total nitrogen, non-protein nitrogen and TCA-precipitable nitrogen were higher, whereas, the ammonia nitrogen and total protozoal count were lower (P<0.01) in herbal product supplemented groups. The results are indicative of supplementation of herbal products enhance the productivity in goats by improving the ruminal environment.

Key words: Herbal products, Marwari goats, Rumen parameters.

INTRODUCTION

Achieving the optimum production at a minimum cost with little or no adverse effects on the environment is the indicative of successful animal production (Wanapat *et al.* 2013). Small ruminants such as goats play an important role in the agricultural systems particularly in the economics of developing countries (Handekar *et al.* 2010). Goats are farmed throughout the world for being important source of milk, meat, fibre as well as of proteins along with their capability of adapting in different environmental conditions (Bhardwaz *et al.*, 2018). The overall performance of ruminants is influenced by the type and amount of feed supplementation (Ngwa *et al.* 2003), depending on the digestibility of feed and rumen fermentation characteristics. An adequate supply of nutrients for livestock is very difficult to the livelihoods of millions of people across the developing world due to large deficiency of feed and fodder. Frequent natural disasters such as droughts and floods further widen the gap between demand and supply of feed and fodder. Due to this reason, animals are mainly fed on such inferior quality feed or by products, which are low in energy and essential nutrients. Therefore, it is desirable to maximize the utilization of nutrients from available feed stuffs either by utilizing complete feed system or by using various feed additives (Manju *et al.* 2013). The growing public awareness over chemical residues in animal products and threats of antibiotic-resistant bacteria have renewed interest in exploring safer alternatives to chemical feed additives in ruminant livestock (Sachan *et al.* 2014). Since synthetic antibiotics as feed additives were banned by the European Union in 2002 (Russell and Houlihan, 2003), scientists have been diverted their interest in evaluating the potential of natural antimicrobials, such as herbs and plant extracts, which have been used for many years for various purposes in human diets. Among non antibiotic feed additives, the

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How to cite this article: Karnani, M., Dhuria, R.K., Sharma, T. and Manju (2021). Effect of Supplementation of Herbal Products as Feed Additive on Rumen Metabolites in Marwari Goats, Indian Journal of Animal Research. 55(2): 185-188. DOI: 10.18805/ijar.B-3932.

Submitted: 30-09-2019 **Accepted:** 21-01-2020 **Online:** 17-03-2020

herbs or herbal preparations have a huge potential to alter ruminal fermentation and improve nutrient utilization in ruminants. Bioactive compounds (*e.g.* thymol, cinnamaldehyde, cineol, methyl eugenol, anethol, allicin *etc.*) present in herbs have been used as immunomodulator to prevent diseases, enhance performance in stress related syndromes and increase resistance against infection (Anjali Kumari *et al.*, 2018). A symbiotic relationship has been found between ruminants and the microbes present in their gut. These microbes in the rumen are responsible for carbohydrates fermentation to produce volatile fatty acids (VFAs), which be responsible for more than half of the total metabolisable energy provided to ruminants. VFAs are weak acids, they disintegrate in the rumen and thus, releasing protons that reduce the pH (Penner *et al.* 2009). Administration of feed additives altered rumen microbial activity which leads to alteration in fermentation process and in turn increase feed utilization efficiency. So, the objective of present study was to ascertain the effect of various herbal products supplementation on the rumen fermentation pattern by estimating various rumen parameters.

MATERIALS AND METHODS

The study was carried out in the Department of Animal Nutrition, College of Veterinary and Animal Sciences, Bikaner. Twenty male Marwari goats of same breed, age group (18-24 months) and of uniform conformation were divided on body weight basis into 5 groups (T_1 , T_2 , T_3 , T_4 and T_5) of four animals in each group. Groups were T_1 (Complete feed without herbal product), T_2 (Complete feed + Himalayan Batisa @ 10 mg/g substrate), T_3 (Complete feed + Appetonic Vet powder @ 10 mg/g substrate), T_4 (Complete feed + Ruchamax @ 3 mg/g substrate) and T_5 (Complete feed + Rumizyme powder @ 7.5 mg/g substrate). The concentration of each herbal product was selected on the basis of their *in vitro* results. Animals were housed in well ventilated, hygienic and protected sheds and were allowed to acclimatize for a period of 10 days prior to experimental feeding. The animals were given prophylactic doses of Fenzole as anthelmintic. The complete feed was prepared with wheat straw as basal roughage source and concentrate in ratio of 60: 40. The animals were given measured quantity of experimental feed and ad lib water every morning.

At the end of feeding trial of 28 rumen liquor sample were collected from each animal at 0, 2, 4, 6, 8, 12 and 18 h post feeding with the help of Ryle's tube introduced through the wooden mouth gag in the rumen against a negative pressure created by the suction pump.

Rumen fluid pH was determined immediately after collection using portable digital pH meter (pen type) at the site of collection. After pH determination one ml of saturated $HgCl_2$ solution was added to each collected sample to kill the microbes and to stop metabolic activity. Rumen liquor was strained through four layers of muslin cloth and referred as strained rumen liquor (SRL). The strained rumen liquor (SRL) was brought to the laboratory in a pre-warmed (39°C) thermos flask and stored in refrigerator. Total volatile fatty acids (TVFA) were determined according to the method of Barnett and Reid (1957) using Markham still distillation apparatus. Total protozoal count of the SRL was done by the method of Moir and Somers (1956) adopted by Purser and Moir (1959) using Sedgewick Rafter Cell (50 × 20 × 1 mm) in 10 × 10 magnification. Rumen ammonia nitrogen in SRL was estimated by Conway's micro diffusion method (1957) using Conway diffusion cell. Total nitrogen was

estimated by Kjeldahl method. 5 ml of rumen liquor was precipitated with 5 ml of 20 per cent trichloroacetic acid and kept overnight in refrigerated condition. Next day after centrifugation at 2000 rpm for 10 minutes, the whole supernatant was transferred in Kjeldahl's digestion tube and nitrogen was estimated. The value thus obtained is reported as non protein nitrogen (NPN) and the difference of above two is reported as TCA precipitable nitrogen. The data obtained in the experiment were analyzed using statistical procedures as suggested by Snedecor and Cochran (1994). Significance of mean differences was tested by Duncan's New Multiple Range Test (DNMRT) as modified by Kramer (1957).

RESULTS AND DISCUSSION

TVFA

Ruminal pH was unaffected by the dietary herbal product supplementation (Table 2). Table 2 showed significant higher values of TVFA concentration were recorded in Himalayan Batisa (T_2) (86.57 mEq/l) and Ruchamax (T_4) (85.65 mEq/l) followed by Appetonic Vet powder (T_3) (83.66 mEq/l) and Rumizyme powder (T_5) (83.57 mEq/l) added groups. The results were in consensus with Singh *et al.* (1992) and Bhatt and Singh (2009) who reported an increase in the TVFA concentration in the strained rumen liquor of animals feeding with different herbal products viz., Himalayan Batisa, Haryana Batisa, Digestovet, Rumbion bolus, Ruchamax and Payapro. The findings were also in agreement with Sardar *et al.* (1997), Manjunatha (1998), Pankaj *et al.* (2008), Ahirwar *et al.* (2009) which might be due to regeneration of VFA producing bacteria by adding polyherbal formulations in the diet.

Total protozoal count

Effect of herbal products supplementation on total protozoal count revealed that supplementation decreased ($p < 0.05$) total number of protozoa in rumen with non significant differences among supplemented groups. Rejil *et al.* (2008) and Amanullah *et al.* (2009) also observed decrease in rumen protozoal number in cattle on feeding *Fenugreek* seed and *Sapindus mukorossi*, *Camelia sinensis* and *Acacia concinna* as feed additive. Ando *et al.* (2003) also found decrease in rumen protozoal numbers in steers fed peppermint as additives.

Table 1: Chemical composition of experimental complete feed and different herbal products (% DM basis).

Attributes	DM	OM	CP	EE	CF	NFE	TA	NDF	ADF	HC	Ca	P
Experimental complete feed	92.80	88.02	14.00	2.47	27.40	44.24	11.98	55.98	35.40	20.98	1.19	0.49
Herbal products												
Himalayan Batisa	97.15	92.59	8.57	1.51	9.47	73.04	7.41	30.46	15.04	15.42	1.74	0.46
Appetonic Vet powder	95.95	93.48	15.23	1.37	10.15	66.73	6.52	54.71	36.96	17.75	1.53	0.58
Ruchamax	96.6	90.5	17.67	1.71	9.86	61.26	9.5	42.63	26.47	16.16	1.61	0.28
Rumizyme powder	97.3	88.64	9.62	1.24	11.14	66.63	11.36	44.19	26.38	17.81	1.29	0.51

Note: Complete feed containing Wheat straw 60%, Deoiled rice bran 16%, Maize 5%, Groundnut cake 4%, Rice bran 4%, Guar korma 4%, Bajra 2%, Molasses 2%, Mineral mixture 2%, Common salt 1%.

Table 2: Rumen fermentation pattern in Marwari goats supplemented with different herbal products.

Diet/Hour	pH	Total protozoal count (x10 ⁵ per ml)	TVFA (meq/L)	NH ₃ -N (mg/dl)	Total N (mg/dl)	TCA-PN (mg/dl)	NPN (mg/dl)
Dietary Treatment							
T1	6.69	3.71 ^b	80.05 ^a	22.52 ^d	88.41 ^a	53.17 ^a	34.77 ^a
T2	6.73	3.33 ^a	86.57 ^c	20.21 ^a	93.55 ^e	57.41 ^d	36.13 ^c
T3	6.74	3.47 ^a	83.66 ^b	21.20 ^c	91.50 ^c	55.98 ^c	35.51 ^b
T4	6.71	3.41 ^a	85.65 ^c	20.74 ^b	92.68 ^d	57.10 ^d	35.57 ^b
T5	6.72	3.53 ^{ab}	83.63 ^b	21.31 ^c	90.26	55.49 ^b	35.23 ^b
	NS	*	**	**	**	**	**
SEM	0.012	0.17	1.00	0.18	0.35	0.42	0.34
Time of Rumen Liquor Sampling (h)							
0 h	6.85 ^f	3.79 ^c	71.05 ^a	19.20 ^b	67.96 ^a	47.47 ^a	20.48 ^a
2 h	6.62 ^b	3.05 ^a	81.17 ^d	21.31 ^d	96.53 ^d	62.25 ^e	34.28 ^c
4 h	6.30 ^a	2.83 ^a	95.21 ^f	24.51 ^f	115.12 ^f	64.48 ^f	50.63 ^g
6 h	6.65 ^c	3.42 ^b	102.51 ^g	22.57 ^e	101.64 ^e	61.38 ^d	40.26 ^f
8 h	6.79 ^d	3.67 ^c	86.50 ^e	21.51 ^d	96.82 ^d	58.89 ^c	37.93 ^e
12 h	6.83 ^e	3.79 ^c	78.44 ^c	20.39 ^c	86.57 ^c	49.25 ^b	37.31 ^d
18 h	6.99 ^g	3.86 ^c	72.51 ^b	18.88 ^a	74.31 ^b	47.10 ^a	27.21 ^b
	**	**	**	**	**	**	**
SEM	0.009	0.13	0.75	0.14	0.26	0.32	0.26

** (P<0.01), NS = Non Significant, N = Nitrogen, TVFA = Total volatile fatty acids, PN = Perceptible Nitrogen, NPN = Non Protein Nitrogen.

Ammonia nitrogen

The concentration of ammonia nitrogen in herbal product supplemented group decreased significantly as compared to control group and indicated reduction in concentration of ammonia nitrogen in rumen liquor on herbal product supplementation (Table 2) with maximum in Himalayan Batisa (T₂) followed by T₄, T₃ and T₅ treatment groups. The reduced level of ammonia nitrogen of herbal product supplementation may not be associated with decreased dietary protein degradation or deamination but appears to be related with increased uptake and assimilation of ammonia nitrogen by rumen microbes due to stimulation of bacterial growth by herbs. Ando *et al.* (2003) and Manjunatha (1998) also observed non-significant rise in ammonia nitrogen on feeding herb as bio stimulator which might be due to faster rate of ammonia metabolism and conversion into microbial proteins. Similarly, Hosoda *et al.* (2006) also reported that clove feeding to Holstein Steers as feed supplement tended (P<0.01) to decrease the ammonia nitrogen concentration compared with those in the control group. Likewise, Bhatt and Singh (2009) also found a decrease in concentration of rumen ammonia nitrogen in Ruchamax and Payapro herbal products supplemented group of cattle.

Total rumen nitrogen

The concentration of total rumen nitrogen obtained for animals fed on various herbal product supplemented and un-supplemented complete feed irrespective of time of sampling were higher values in herbal products supplemented

groups as compared to control group (Table 2). Among supplemented groups T₂ showed significantly highest concentration of ruminal nitrogen and T₅ showed lowest concentration.

TCA-precipitable nitrogen

Table 2 showed that TCA-precipitable nitrogen concentration was significantly higher in herbal products supplemented groups. This might be due to increased utilization of ammonia nitrogen by rumen microbes for microbial protein synthesis. Bhatt and Singh (2009) also observed an increase in the concentration of TCA precipitable nitrogen in animals feeding with Ruchamax and Payapro.

Non protein nitrogen

A non significant effect in non protein nitrogen concentration on supplementation of herbal products was shown in Table 2. Rani *et al.* (2006) also reported similar findings of no effect on non protein nitrogen concentration on supplementation of herb.

CONCLUSION

Based on the results of present study, it may be concluded that supplementation of herbal products *viz.*, Himalayan Batisa, Appetonic Vet powder, Ruchamax and Rumizyme powder as a feed additive have beneficial effect on rumen fermentation pattern of goat. It is revealed that supplementation of herbal products can also be used as a part of strategy to be adopted to improve productivity of goats in arid and semi-arid region and to have lucrative goat farming.

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